One-Sample Mean Inference (Solutions)

Testing a population mean

Parameter of interest: μ

- Point estimate: $\bar{X}$
- Standard error: $\frac{\sigma}{\sqrt{n}}$
- Model: T_{n-1}

Let’s return to the penguin data from the start of the semester.

What is the population average flipper length of Adelie penguins?



What reference distribution should we use for mean inference?

A simulation of the sample mean shows a bell-curve shape for large enough n.

The Central Limit Theorem says that for large enough n,

$$\bar{X} \dot{\sim} N\left(\mu, \frac{\sigma}{\sqrt{n}}\right).$$

If we apply standardization, we get

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \dot{\sim} N(0, 1)$$

Should we use a Z-test with test statistic $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$?



Problem: we don't know the value of σ, the population standard deviation. This adds extra error.

Let's substitute σ with the *sample* standard deviation S.

$$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

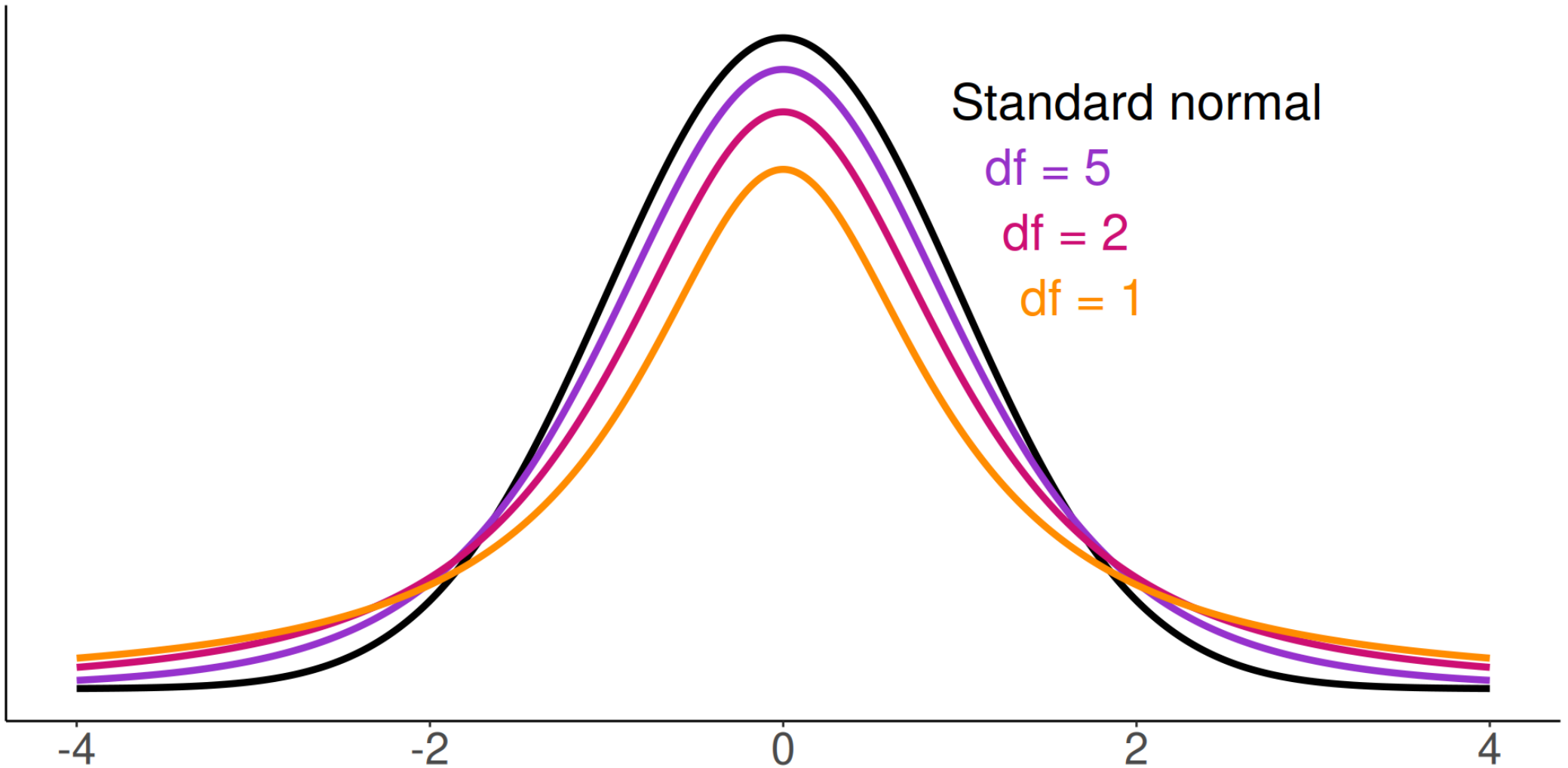
normal

$$\frac{\bar{X} - \mu}{S / \sqrt{n}}$$

$\text{not *quite* normal}$

The right term follows a **Student's T distribution**.

The T curve is wider, representing the additional error from having to estimate σ.



The extra error is controlled by the **degrees of freedom**.

In one-sample mean inference, $df = n - 1$.

Bigger n/df	**Smaller** n/df
S similar to σ	S different from σ
Less error	More error
Narrower T curve	Wider T curve

Use R functions `pt()` and `qt()`.

Exercises

Let Z have a $N(0,1)$ distribution, T_5 have a T distribution with 5 df, and let T_{10} have a T distribution with 10 df.

- Find $P(X < 0)$ for each of the three variables.
- Find $P(X < -1)$ for each of the three variables.
- Find the 40th percentile for each of the three variables.
- CHALLENGE: How big must the df be before the 40th percentile of a T distribution is within 0.005 of the 40th percentile of Z?



Solutions/Hints

- $P(X < 0)$ is exactly 0.5 for all three variables, since both the normal and T are symmetric around 0.
- $P(X < 0)$ is largest for T_5 since the T distribution (with lower df) has more area in the tails.
- The 40th percentile for T_5 is smallest, i.e. furthest from 0. The 40th percentile of T_{10} is closer to the 40th percentile of Z.
- The 40th percentile of T_{14} is within 0.005 of the 40th percentile of Z.

Formal hypothesis testing steps (T test):

1. State hypotheses and parameter of interest (μ)
2. Identify null distribution + assumptions (T distribution)
3. Calculate observed test statistic + p-value (t_{obs}, `pt()`)
4. Make a conclusion in context (about μ)

Let's perform a one-sample T test on the Adelie penguin data, to test whether the average flipper length is 192 mm. Let's use $\alpha = 0.01$.

$$H_0 : \mu = 192 \quad \text{versus} \quad H_A : \mu \neq 192$$

Are the conditions for this test well-met?

Our test is based on the CLT. We need:

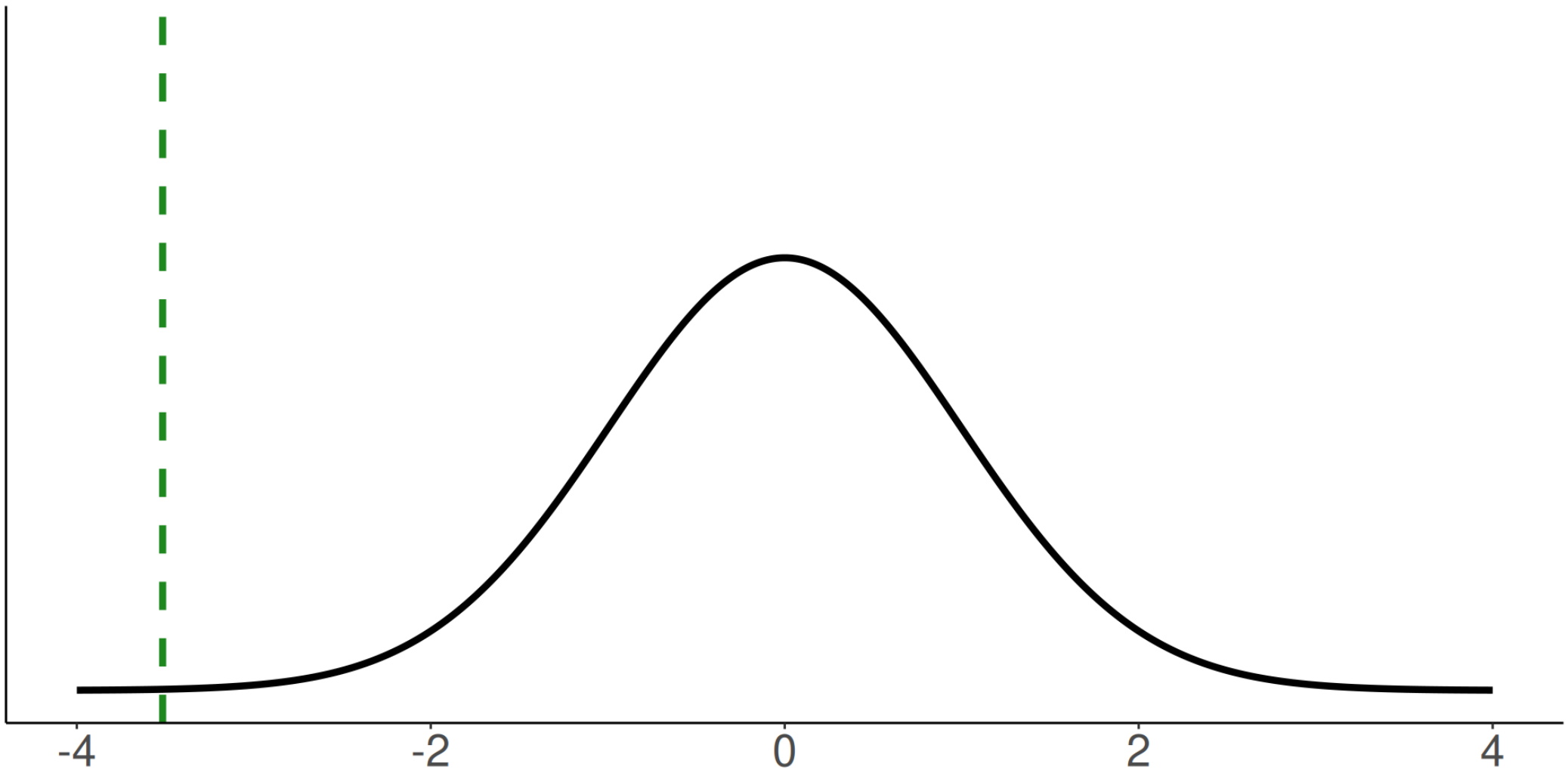
- A large sample and/or approximately normal data
- Independent draws from the same population

We have observed test statistic

$$t_{obs} = \frac{\bar{x} - 192}{s/\sqrt{n}} = -3.515$$

which has $n - 1 = 145$ degrees of freedom. Our p-value is calculated on the t distribution.

Our test statistic $t_{obs} = -3.515$ is far from 0, in the lower tail.



The p-value is the area below -3.515 plus the area above 3.515.

Using `pt()`, we get a p-value of about 0.0006, which is less than our significance level $\alpha = 0.01$.

We can also check our work with `t.test()`.

In general, we can test hypotheses

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_A : \mu \neq \mu_0$$

with test statistic

$$t_{obs} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

and a null t_{n-1} distribution. This works if we have independent observations that look approximately normal.

What if we instead performed a Z test?

The test statistic calculation is the same, but we calculate a p-value with `pnorm()`.

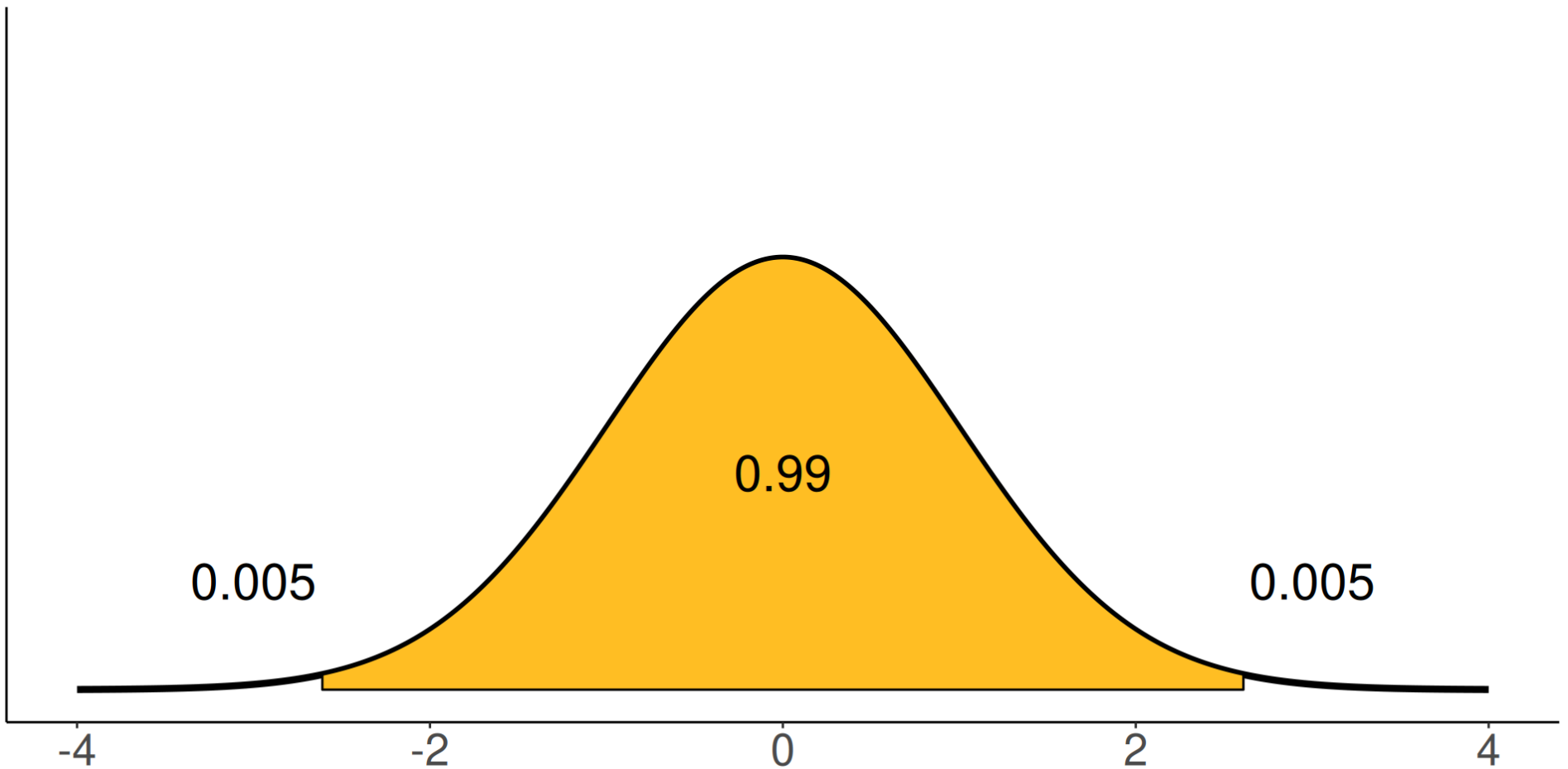
The result is similar, since T_{145} is very similar to $N(0, 1)$.

We can also build a T CI for a population mean μ. The general CI formula is

$$\text{point estimate} \pm \text{critical value} \times \text{standard error}$$

For μ, the point estimate is $\bar{X}$, the estimated standard error is $\frac{S}{\sqrt{n}}$, and we use a T critical value.

Critical values for a 99% T CI:



this picture looks exactly like one we've seen before but just trust me it is actually a T with 145 df and not a standard normal

Our CI for mean flipper length is

$$190.1 \pm 1.61 \times \frac{6.522}{\sqrt{146}} = (188.69, 191.51)$$

Notice how 192 is not in the interval!

The one-sample T CI is given by

$$\bar{X} \pm t_{n-1,\alpha/2} \times \frac{S}{\sqrt{n}}$$

where $t_{n-1,\alpha/2}$ is the $\alpha/2$ critical value on the t curve with $n-1$ degrees of freedom.

Exercises

Perform a T test and build a T CI from scratch! Use $\alpha = 0.02$. We have a dataset of $n = 15$ masses of ball bearings manufactured at a facility. The weight should be calibrated to $\mu = 12$.

- Graph the data to see if normality holds.
- Set up hypotheses about μ.
- Calculate a T test statistic and p-value.
- Build a 98% CI for μ.



Solutions/Hints

- We are testing $H_0 : \mu = 12$ versus $H_A : \mu \neq 12$.
- The observed test statistic is about -1.81, leading to a p-value of 0.092. We fail to reject the null.
- The 98% CI contains the value 12.

The following 2 statements are equivalent for T CIs and hypothesis tests for μ:

- A $100(1-\alpha)\%$ CI does contains μ_0.
- A test for $H_A : \mu \neq \mu_0$ at level α fails to reject H_0.

Conversely, if μ_0 is outside the interval, the hypothesis test will reject H_0.

Looking ahead

In chapter 13, we'll extend the T test to the problem of studying a difference in population means. If you want a head start, consider taking samples from two different populations:

Mean	SD	Sample size
$\bar{X}_1$	S_1	n_1
$\bar{X}_2$	S_2	n_2

The standard error of $\bar{X}_1 - \bar{X}_2$ is $\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$. How does this expression simplify if $S_1 = S_2 = S$? What would we use for S if S_1 and S_2 are not exactly equal?